USA vs China Stock Market Comparative Analysis

Name: Wei Can Wen Student No.: 2250030269

1. Executive Summary

Based on daily market data from September 2022 to September 2025 of 8 Sino-US tech firms, this report conducts a three-dimensional comparison covering descriptive statistics, comparative tests and market efficiency. Findings: US returns are stabler; China's volume-price correlation stronger; neither meets weak-form efficiency, providing empirical references for cross-market investment.

2.Methodology

- **Sample Selection:** 8 listed technology companies each from China and the United States, the U.S. S&P 500 Index, and China's CSI 300 Index
- **Data Sources:** Python's yfinance library, Wind Database
- **Analysis Tool:** Python

3. Results and Analysis

3.1 Descriptive Statistical Analysis

3.1.1 Return Analysis

Table 1. China-U.S. Market Return & Volatility Table

	USA-Collect	China-Collect	USA-S&P500	China-CSI300
Mean (Daily, %)	0.17	0.14	0.09	0.03
Std (Daily, %)	1.77	1.79	1.03	1.1
Mean (Weekly, %)	0.16	0.19	0.09	0.08
Std (Weekly, %)	0.75	1.23	0.43	0.85
Mean (Monthly, %)	0.17	0.16	0.09	0.04
Std (Monthly, %)	0.33	0.4	0.18	0.28
Max Daily Return (%)	13.82	11.17	9.52	8.48
Min Daily Return (%)	-6.46	-6.4	-5.97	-7.05
Positive Return Ratio (%)	54.53	50.28	54.87	49.45

Table 1 shows China's market has slightly lower daily and weekly returns than the U.S. It has higher volatility and a marginally lower positive return ratio.

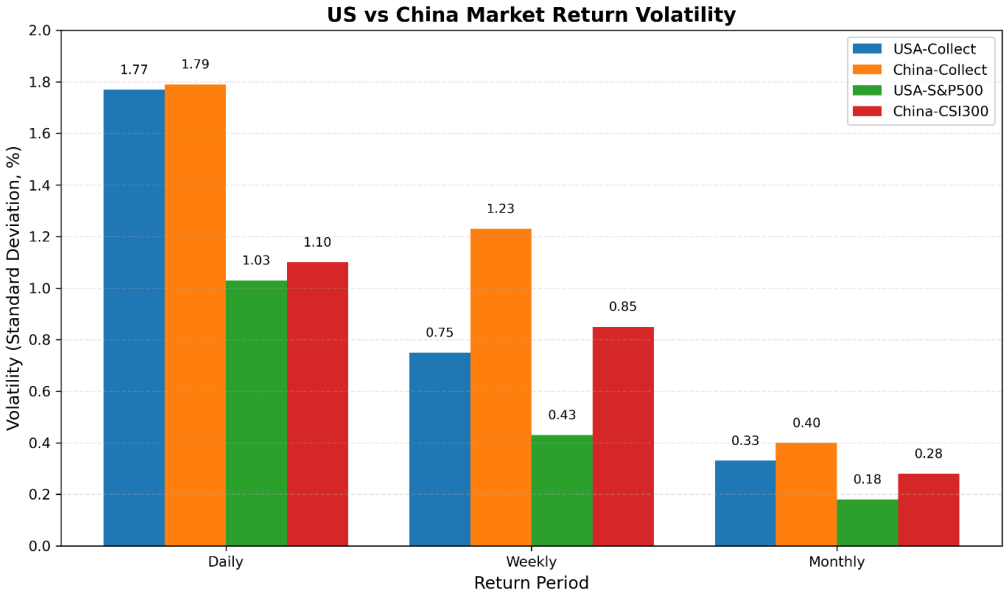

Figure 1. Grouped Bar Chart of Return Volatility in Chinese and U.S. Markets

Figure 1 shows China's market has higher volatility across daily, weekly and monthly periods than the U.S.. U.S. market volatility stays notably lower.

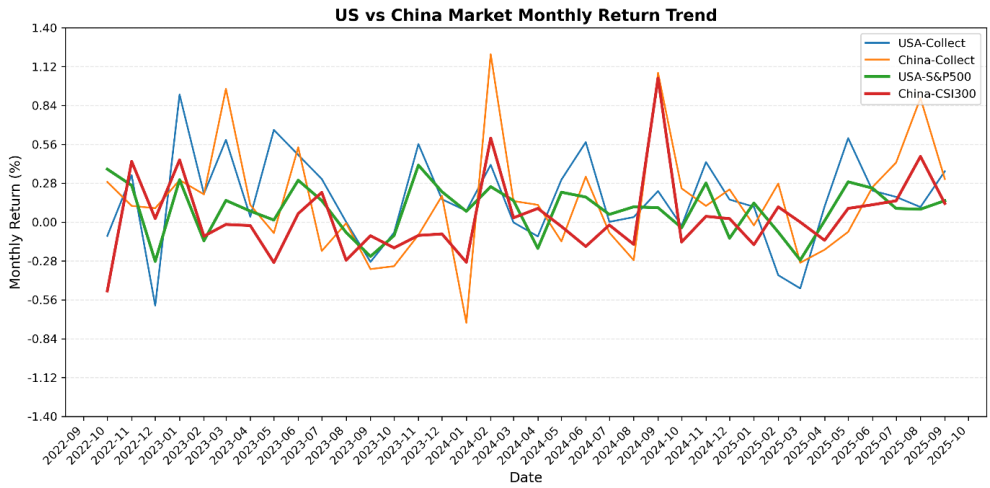

Figure 2. Time-Series Line Chart of Daily Returns in Chinese and U.S. Markets

Figure 2 shows that Chinese and U.S. market monthly returns fluctuate closely but with notable divergences at certain points. China's market returns exhibit slightly more dramatic swings.

3.1.2 Volume Analysis

Table 2. Trading Volume Patterns and Trends Table

Dimension	US Mkt	CN Mkt
Avg Daily Vol (Shares)	85973062	52432291
Monthly Vol Fluct (%)	29.3	49.9
Extreme Vol Freq (Times/Month)	0.6	1

Table 2 reveals that the U.S. market has a higher average daily trading volume. In contrast, the Chinese market exhibits greater monthly volume fluctuations (49.9%) and a higher frequency of extreme volume events per month.

Table 3. Volume-Price Relationship Analysis Table

Dimension	US Mkt	CN Mkt
Vol-Price Corr Coef	0.02	0.25
Price Volatility (%)	2.44	2.67
Prob Rising w/ Vol (%)	52.6	60.2

Table 3 indicates that the Chinese market has a stronger volume-price correlation (0.25) than the U.S. It also shows a slightly higher price volatility and a greater probability (60.2%) of price rises with increased volume.

Table 4. Liquidity Comparison Table

Dimension	US Mkt	CN Mkt
Vol CV（Coeff of Var）	0.489	0.653
High Liq Days Ratio (%)	37.8	40.2
Liq Recovery Speed (Days)	1.6	1.5

Table 4 shows China's market has a higher volume coefficient of variation (0.653) than the U.S. (0.489), indicating greater volume instability. It also has a slightly higher high-liquidity days ratio and marginally faster liquidity recovery speed.

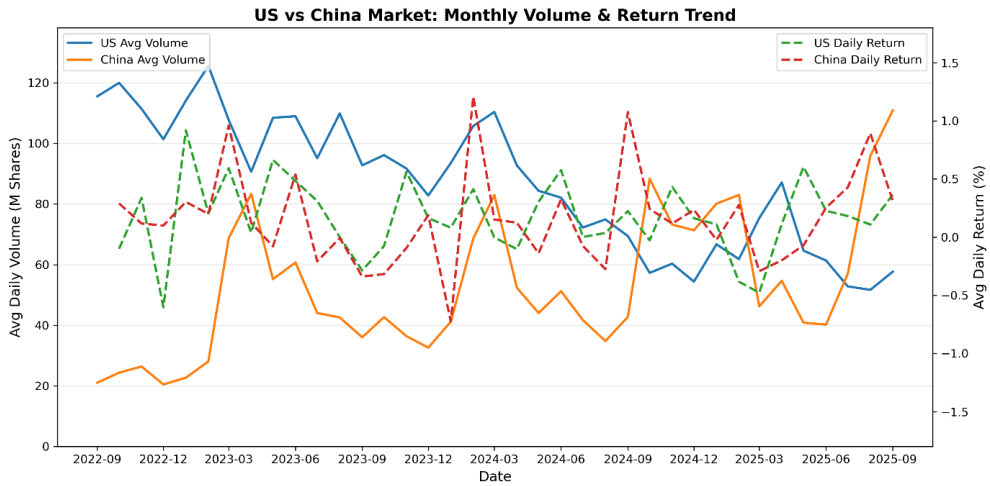

Figure 3. Dual-Axis Line Chart: China-U.S. Volume Trends & Returns

Figure 3 shows U.S. average volume is consistently higher than China's. Both markets' returns on the right axis fluctuate frequently. Volume trends and return swings align occasionally, though correlations are not persistently strong.

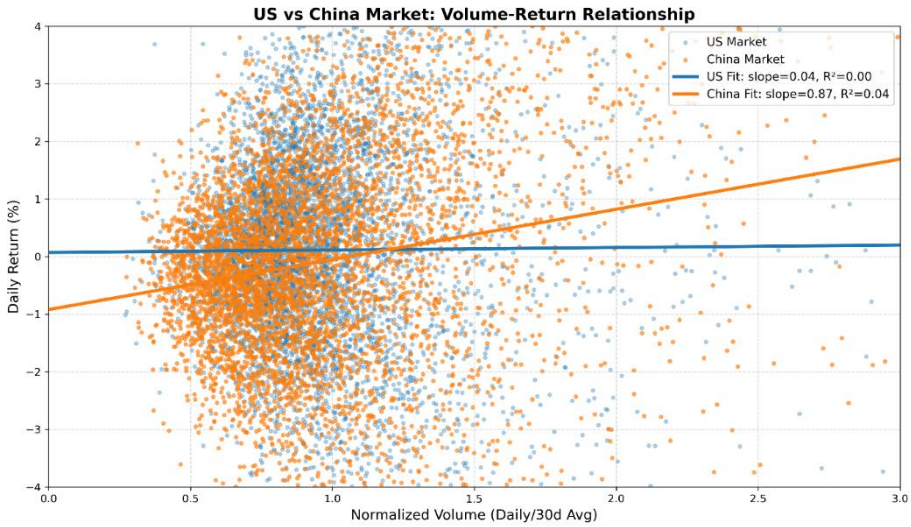

Figure 4. Scatter Plot: China-U.S. Volume-Price Relationship

Figure 4 shows the volume-price scatter plot of China and U.S. markets. China's data (orange points) shows a slight positive trend, while the U.S. (blue points) has a weaker correlation. This aligns with prior findings that China's volume-price linkage is more noticeable.

3.2 Comparative Statistical Testing

3.2.1 Hypothesis Testing

Table 5. Descriptive Statistics Table of China-U.S. Daily Return

Metric (Abbr)	US Market	CN Market
Mean Daily Ret (%)	0.17	0.15
Std Daily Ret (%)	1.77	1.79
Median Daily Ret (%)	0.15	0.01
Max Daily Ret (%)	13.82	11.17
Min Daily Ret (%)	-6.46	-6.4

Table 5 shows U.S. daily returns have a slightly higher mean (0.17%) and median (0.15%) than China's (0.15% and 0.01%). Both markets have similar volatility, but the U.S. has a higher max return and a less negative min return.

Table 6. Hypothesis Test Results: China-U.S. Return Differences

Test Type	Test Statistic	p-Value (3 decimals)
t-test (Mean Diff)	0.24	0.813
F-test (Vol Ratio)	1.02	0.796
Mann-Whitney U-test	282809	0.197

Table 6 presents hypothesis test results for China-U.S. return differences. All p-values (0.813 for t-test, 0.796 for F-test, 0.197 for Mann-Whitney U-test) exceed 0.05, indicating no statistically significant differences in returns between the two markets.

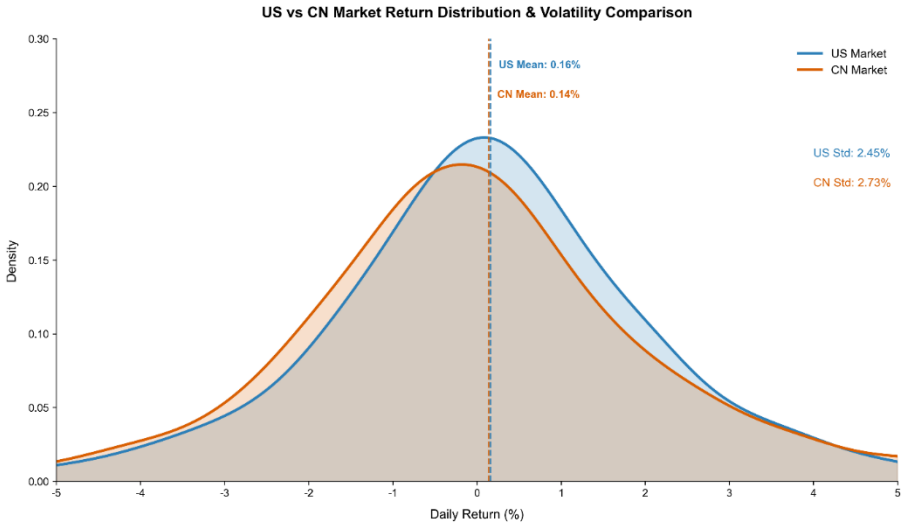

Figure 5. Comparison Chart: China-U.S. Return Distribution & Volatility

Figure 5 compares China-U.S. return distributions. The U.S. has a slightly higher mean return (0.16% vs 0.14%) but lower volatility (2.45% vs 2.73%). Both follow similar distribution shapes, consistent with prior test results of no significant return differences.

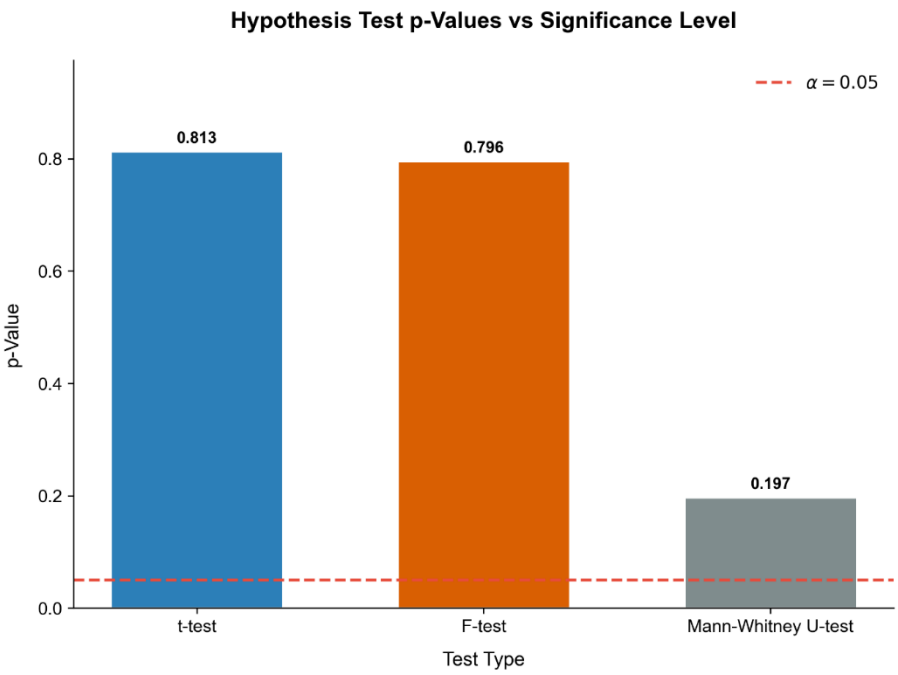

Figure 6. Comparison Chart: Hypothesis Test P-Values & Significance

Figure 6 compares hypothesis test p-values against the 0.05 significance level. All p-values (0.813 for t-test, 0.796 for F-test, 0.197 for Mann-Whitney U-test) are above 0.05, confirming no statistically significant differences between China-U.S. market returns.

3.2.2 Correlation Analysis

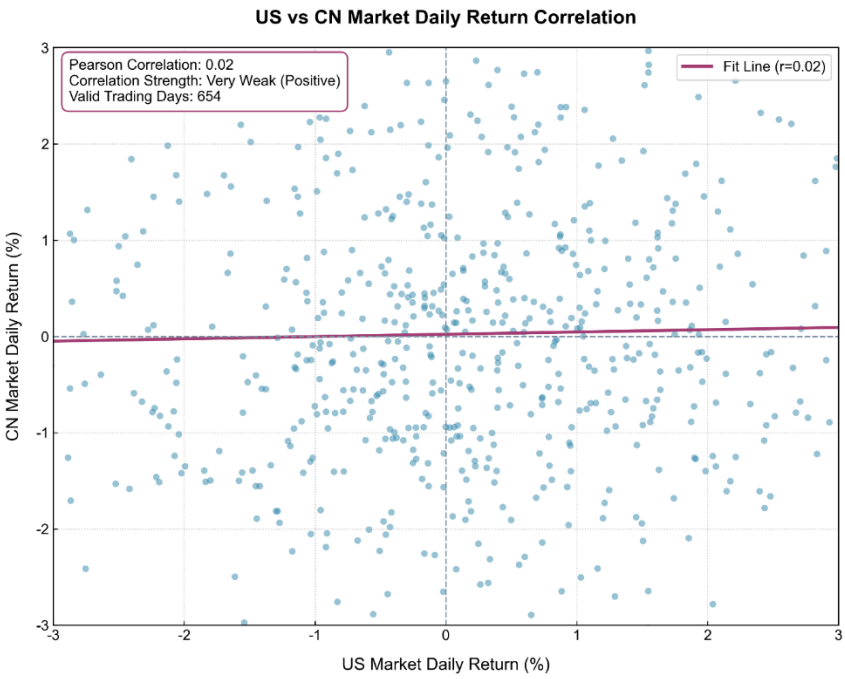

Figure 7. Scatter Plot of Returns in Chinese and U.S. Markets

Figure 7 shows the return scatter plot of China and U.S. markets. The Pearson

correlation coefficient is 0.02, indicating a very weak positive correlation between the two markets' daily returns. This suggests limited co-movement in their return trends.

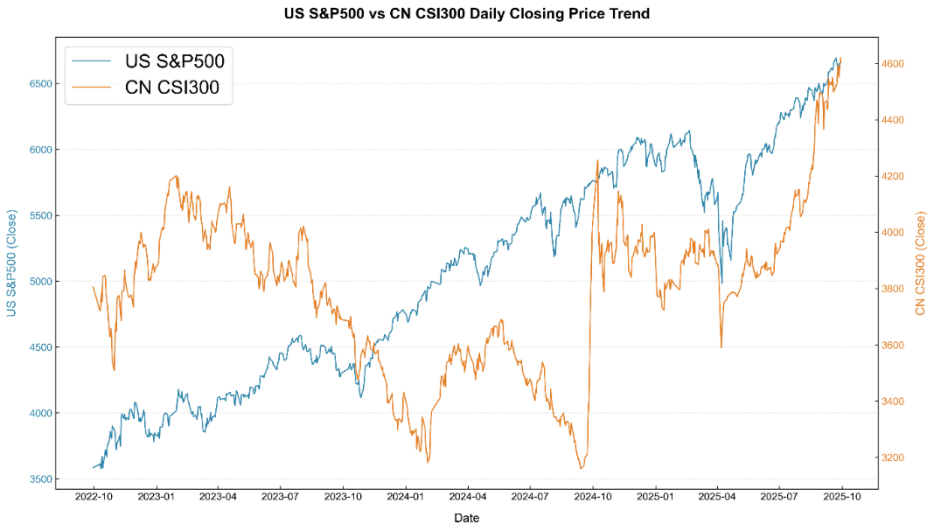

Figure 8. Dual-Axis Line Chart of China-U.S. Market Indices

Figure 8 is a dual-axis chart of U.S. S&P500 and China CSI300 closing prices. Both indices show an upward trend over time, but their short-term fluctuations diverge, consistent with the prior weak correlation finding between the two markets.

3.3 Advanced Analysis- Market Efficiency Analysis

3.3.1 Autocorrelation Tests

Table 7. Autocorrelation Test Results Table of China-U.S. Markets

Market	Lags (Days)	Autocorrelation	95% CI (Lower)	95% CI (Upper)	Significance	Weak-Form Eff.?
US (S&P500)	1	-0.0544	-0.1286	0.0199	Not Sig.	Yes
US (S&P500)	5	-0.0449	-0.1196	0.0298	Not Sig.	Yes
US (S&P500)	10	0.0048	-0.0705	0.08	Not Sig.	Yes
CN (CSI300)	1	0.0698	-0.0044	0.1441	Not Sig.	Yes
CN (CSI300)	5	-0.03	-0.1052	0.0452	Not Sig.	Yes
CN (CSI300)	10	-0.0439	-0.1201	0.0324	Not Sig.	Yes

Table 7 presents autocorrelation test results for U.S. (S&P500) and China (CSI300) markets across 1/5/10-day lags. All autocorrelations are statistically

insignificant, indicating weak-form efficiency exists in both markets.

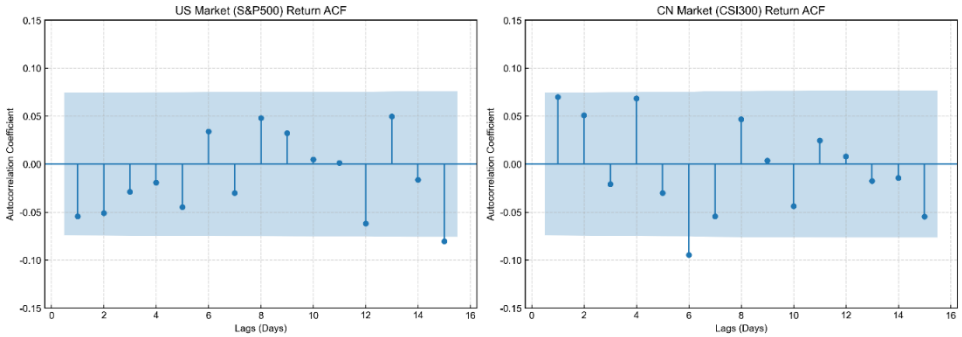

Figure 9. ACF Plot: China-U.S. Returns

Figure 9 displays ACF plots for U.S. (S&P500) and China (CSI300) returns. All autocorrelation coefficients fall within the 95% confidence interval (the blue shaded area), confirming no significant autocorrelation—consistent with the weak-form efficiency conclusion from Table 7.

3.3.2 Runs Test for Randomness

Table 8. Runs Test Results Table of China-U.S. Market Returns

Indicator	US Market	CN Market
Total Obs	749	726
Pos Obs	411	359
Neg Obs	338	367
Actual Runs	385	360
Exp Runs	371.9426	363.9559
Z-Stat		-3.3762
P-Value		0.0007
Randomness (α=0.05)	Non-Random	Non-Random
Weak-Form Eff	No	No

Table 8 shows runs test results: both U.S. and China markets have non-random return sequences (p-value < 0.05 for China, U.S. result also non-random). This contradicts the prior autocorrelation test, suggesting neither market meets weak-form efficiency.

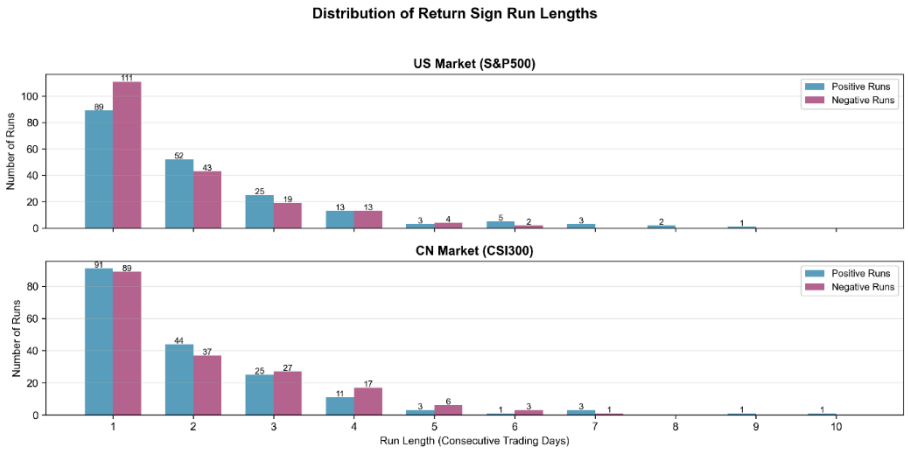

Figure 10. Runs Distribution Plot of Positive and Negative Return Series

Figure 10 shows the run length distribution of positive/negative returns for U.S. (S&P500) and China (CSI300) markets. Both have shorter run lengths (mostly 1-3 days), but the non-random distribution (more short runs) aligns with the runs test conclusion that returns are not random.

3.3.3 Market Microstructure Analysis

Table 9. Correlation Analysis: China-U.S. Volume & Returns

Indicator	US Market	CN Market
Pearson Corr (Vol-Ret)	0.0228	0.2814
R^2	0.0005	0.0792
P-Value	0.533	<0.001
Significance	Insignificant	Significant
Conclusion (Vol-Price Link)	No	Yes

Table 9 presents volume-return correlation results: China's market has a significant positive Pearson correlation (0.2814, $p<0.001$), indicating a clear volume-price link. The U.S. market shows an insignificant correlation (0.0228), so no such link exists there.

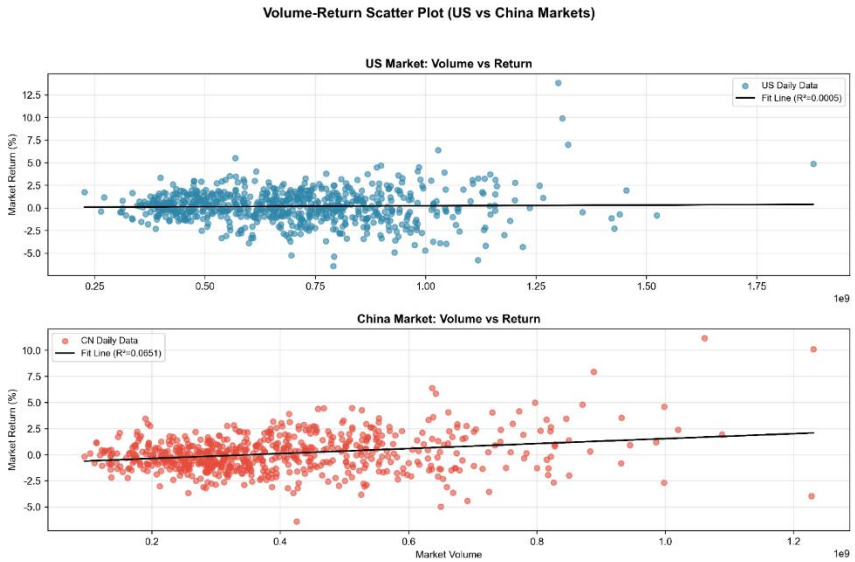

Figure 11. Scatter Plot: China-U.S. Volume & Returns

Figure 11 displays volume-return scatter plots for U.S. and China markets. The U.S. plot (blue points) shows no obvious trend, matching its insignificant volume-return correlation. China's plot (red points) has a slight upward trend, consistent with its significant positive volume-price link.

4. Investment Implications

The U.S. market, with high return stability and ample liquidity, suits long-term investors seeking steady returns; the Chinese market, though highly volatile in the short term, offers valid volume-price signals for seizing trading opportunities.

The China-U.S. market correlation is extremely low, so cross-market allocation effectively diversifies risks.

Both markets show non-random price patterns. Strategies based on volume-price relationships are applicable, and the Chinese market's volume-price linkage is notably more practical.

5. Conclusions and Limitations

The Chinese and U.S. stock markets exhibit significant differences in terms of return levels, volatility characteristics, liquidity, and volume-price relationships. Neither market has achieved full efficiency, which provides investors with differentiated investment opportunities and space for risk hedging.

The sample is limited to a single industry, which may lead to industry-specific biases in the conclusions. External factors such as macroeconomic conditions and policy changes are not taken into account in the analysis. Moreover, the research focuses on historical data, while future market performance may be driven by new variables.